

Experiment No.03

HTML CODE:

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Calculator</title>
  <link rel="stylesheet" href="cal.css" />
</head>

<body>
  <div id="container">
    <div id="card">

      <h1>Calculator App</h1>

      <p>
        Designed by
        <a href="https://github.com/gauravpatrekar01">
          Gaurav Patrekar
        </a>
      </p>

      <div>
        <input
          type="text"
          id="nums"
          placeholder="Enter the Numbers"
        />
      </div>

      <div class="btns">

        <button id="add" class="num-btn" onclick="btnAdd()">+</button>

        <button id="sub" class="num-btn" onclick="btnSub()">-</button>

        <button
          id="mul"
```

```
        class="num-btn"
        onclick="btnMul()"
        style="background-color: #4CAF50; color: white;"
    >*</button>

<button
    id="div"
    class="num-btn"
    onclick="btnDiv()"
    style="background-color: #2196F3; color: white;"
>/</button>

<button id="Equal" class="num-btn" onclick="btnEqual()">=</button>

<br />

<button class="num-btn" onclick="appendNum('1')">1</button>
<button class="num-btn" onclick="appendNum('2')">2</button>
<button class="num-btn" onclick="appendNum('3')">3</button>

<br />

<button class="num-btn" onclick="appendNum('4')">4</button>
<button class="num-btn" onclick="appendNum('5')">5</button>
<button class="num-btn" onclick="appendNum('6')">6</button>

<br />

<button class="num-btn" onclick="appendNum('7')">7</button>
<button class="num-btn" onclick="appendNum('8')">8</button>
<button class="num-btn" onclick="appendNum('9')">9</button>

<br />

<button class="num-btn" onclick="appendNum('0')">0</button>
<button class="num-btn" onclick="btn00()">00</button>
<button id="clear" class="num-btn" onclick="clearInput()">C</button>

</div>
</div>
</div>
```

```
<script src="cal.js"></script>
</body>
</html>
```

JS CODE:

```
console.log("Load cal.js");

function appendNum(num) {
    var numsInput = document.getElementById("nums");
    numsInput.value += num;
}

function btn00() {
    appendNum(0);
    appendNum(0);
}

function btnAdd() {
    appendNum("+");
}

function btnSub() {
    appendNum("-");
}

function btnMul() {
    appendNum("*");
}

function btnDiv() {
    appendNum("/");
}

function btnEqual() {
    var numsInput = document.getElementById("nums");

    try {
        var expression = numsInput.value;

        if (expression === "") {
```

```
        throw new Error("Input cannot be empty");
    }

    if (!/^[0-9+\-*/.\s]+$/.test(expression)) {
        throw new Error("Invalid characters in input");
    }

    if (/^[+\-*/]$/.test(expression.trim())) {
        throw new Error("Incomplete expression");
    }

    var result = eval(expression);

    if (!Number.isFinite(result)) {
        throw new Error("Invalid calculation");
    }

    numsInput.value = result;

} catch (error) {
    console.error("Error:", error.message);
    alert(error.message);
}

}

function clearInput() {
    var numsInput = document.getElementById("nums");

    try {
        if (numsInput.value === "") {
            throw new Error("Input is already empty");
        }

        numsInput.value = "";

    } catch (error) {
        console.error("Error:", error.message);
        alert(error.message);
    }
}
```

CSS CODE:

```
*{box-sizing:border-box;font-family:Arial,Helvetica,sans-serif}
body{height:100vh;margin:0;display:flex;justify-content:center;align-
items:center;background:linear-gradient(135deg,#84ac25,#6709af)}
#card{width:340px;padding:30px 25px;background:white;border-radius:20px;box-
shadow:0 15px 40px rgba(0,0,0,.25);text-align:center}
h1{margin:0 0 5px;font-size:30px;color:#222}
p{margin:5px 0 20px;font-size:14px;color:#666}
p a{color:#6709af;text-decoration:none}p a:hover{text-decoration:underline}
input[type="text"]{width:100%;max-width:270px;height:55px;margin:0 auto
20px;padding:10px 15px;display:block;font-size:15px;text-align:center;border:2px solid
#ddd;border-radius:10px;outline:none;background:#f8f8f8}
input[type="text"]:hover{border-color:#6709af;background:white}
.btns{display:flex;flex-wrap:wrap;justify-content:center}
.num-btn{width:55px;height:55px;margin:5px;border:0;border-radius:10px;font-
size:20px;font-
weight:bold;background:rgb(20,47,51);color:white;cursor:pointer;transition:.2s}
.num-btn:hover{background:#0b784c;transform:translateY(-2px)}
.num-btn:active{transform:scale(.95)}
#add,#sub,#mul,#div{background:#6709af}#add:hover,#sub:hover,#mul:hover,#div:hov
er{background:#4d067f}
#Equal{background:#84ac25}#Equal:hover{background:#668b18}
#clear{width:115px;background:#d62828}#clear:hover{background:#a91f1f}
@media(max-width:400px){#card{width:90%;padding:25px 15px}.num-
btn{width:50px;height:50px;margin:4px}input[type="text"]{width:100%}}
```

OUTPUT:



